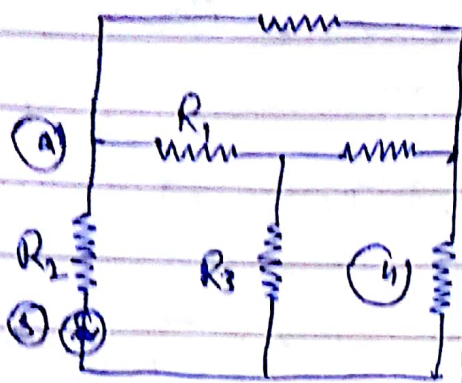


GRAPH THEORY



method to solve electrical
Given - any electrical network

graph

oriented graph

tree/co-tree

tie set matrix

cut set matrix

Tree - It contains all the nodes of a graph.

- i) There is no closed path and hence tree is circuit less.
- ii) If the graph contains n nodes, tree will have $(n-1)$ branches.
- iv) Tree can be used to solve the electrical network using tie set matrix and cut set matrix.

Tree - Branches - It is any branch of a tree also called twig.

Tree link (chord). It is that branch of a graph that does not belong to a particular tree called co tree, it is simply called as links.

The no. of branches in a tree $(n-1)$, no of link (chord) $b-(n-1)$

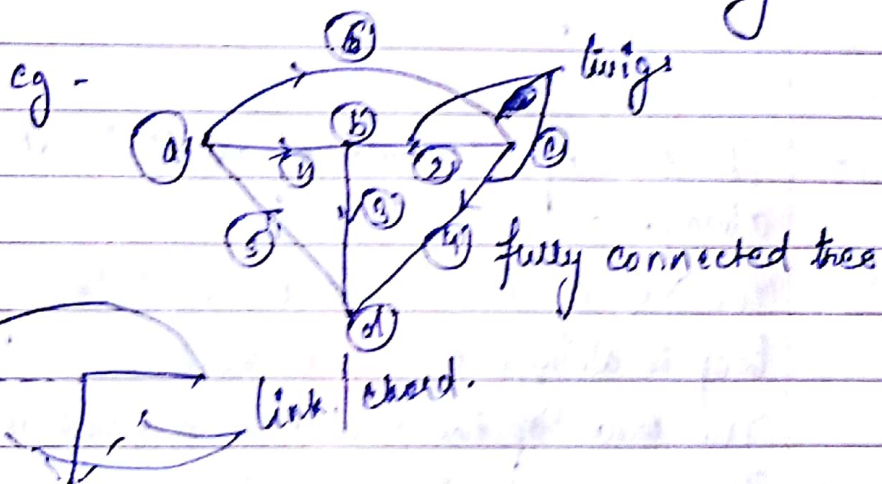
b total no. of branches. (ie no of branches + links in a tree)

- formula to find the no. of trees -

$\det[A A^T]$ - this formula is valid for both fully connected and not fully connected graph.
Incidence matrix

no. of trees in a fully connected graph $n^{(n-2)}$

Fully connected graph - Here all the nodes are connected to each other node directly.



a ~~part~~ tree from the same network.

The tree is defined as any set of branches that is just sufficient to connect all the nodes.

The no. of tree branches/twigs $2n-1$

The branches of the graph not in tree form co-tree.

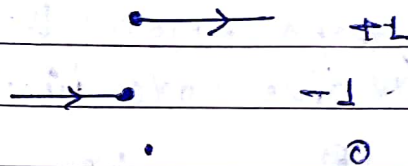
The link is any branch belonging to co-tree.

Graph is the union tree and its co-tree. The

decomposition of graph in tree / co-tree or decomposition of branches into twigs and links is not unique.

Incidence matrix

- This matrix translates all the geometrical features of a graph into an algebraic expression.
- Each graph has an incidence matrix and vice versa.
- Each row contains an entry of $+1$, -1 or 0 depending upon which branch is connected to a particular node.



- The sum of elements of individual column is always zero.
- The det of the incidence matrix of a closed loop is always equal to zero.
- The order of the incidence matrix is $(n \times b)$
- Two graphs having same incidence matrix is called isomorphic graph.
- The rank of incidence matrix of a fully connected graph $= (n-1)$

		Branches					
		1	2	3	4	5	6
Nodes	a	$+1$	0	0	0	$+1$	$+1$
	b	-1	-1	$+1$	0	0	0
	c	0	$+1$	0	$+1$	0	-1
	d	0	0	-1	-1	$+1$	0

Incidence matrix

Reduced Incidence Matrix -

A reference node is selected and a row corresponding to that node is deleted in the Incidence matrix then the remaining part is called reduced incidence matrix.

Degree -

It is no. of branches incident to it.

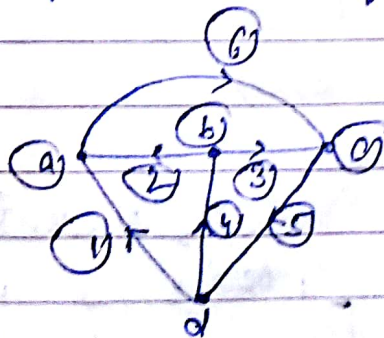
Rank -

Max. no. of branches incident to any node.

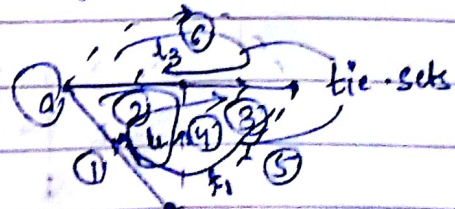
Degree = Rank (for fully connected graph).

Tie sets -

- A fundamental tie set represents a set of branches which form a closed loop.
- Each fundamental tie set contains one and only one link/chord, remaining are all tree/branches.
- The direction of each fundamental tie set is same as the direction of link/chord.
- No. of tie sets = no. of links/chords = $b - (n - 1)$



Taking (a) [1, 2, 3] as tree



- Tie set is formed like a KVL equation with one link/chord and rest branches in a loop taken.

$l_1 = [1, 2, 3, 5]$
 $l_2 = [1, 2, 4]$
 $l_3 = [2, 3, 6]$

→ link / chord.

	1	2	3	4	5	6
l_1	+1	-1	+1	0	+1	0
l_2	-1	+1	0	+1	0	0
l_3	0	+1	-1	0	0	+1

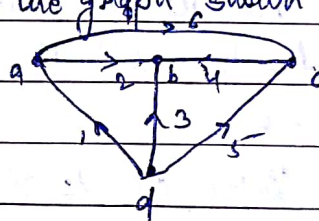
$[B_f] \rightarrow$

	1	2	3	4	5	6
l_1	-1	+1	0	1	0	0
l_2	+1	-1	+1	0	1	0
l_3	0	+1	-1	0	0	1

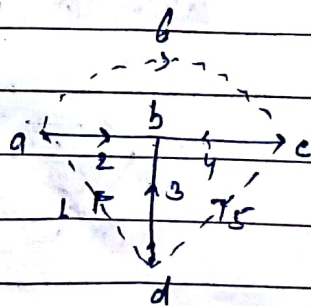
link / chord is taken in increasing order.

unity matrix

Q → For the graph shown in figure consider the tree formed by branches $[2, 3, 4]$. Using the tree write.



- Incidence matrix A .
- B_f
- Q_f .



$l_1 = [1, 2, 3, 5]$
 $l_2 = [1, 2, 4]$
 $l_3 = [2, 3, 6]$

	1	2	3	4	5	6
l_1	-1	+1	0	0	0	+1
l_2	0	-1	-1	-1	0	0
l_3	0	0	0	+1	-1	-1
l_4	+1	0	+1	0	+1	0

$$l_1 = [1, 2, 3, 4, 5]$$

$$l_2 = [4, 3, 5]$$

$$l_3 = [4, 2, 6]$$

	2	3	4	1	5	6
l_1	+1	-1	0	+1	0	0
l_2	0	-1	+1	0	+1	0
l_3	-1	0	+1	0	0	+1

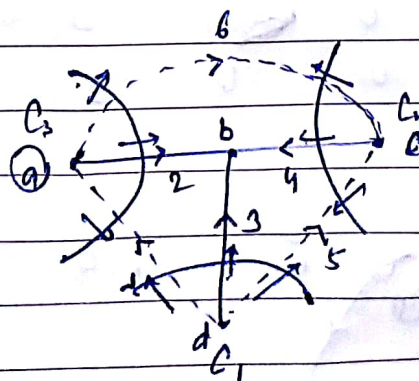
$$\left[B_f \mid U \right]$$

from tree $\swarrow$ $\searrow$ unity

Cut Set matrix

- Each fundamental cut set is a set of branches which must be cut so that a graph is divided into two parts.
- Each fundamental cut set contains only one branch, remaining are all links/cords.
- Direction of each fundamental cut set is same as that of a tree/branch.

- no. of f-cut sets $= n - 1$



branch $\uparrow$

$$C_1 = [1, 3, 5]$$

$$C_2 = [6, 4, 5]$$

$$C_3 = [1, 6, 2]$$

	1	2	3	4	5	6
e_1	-1	0	+1	+1	0	0
e_2	+1	+1	0	0	+1	0
e_3	0	-1	-1	0	0	+1

$$[Q_1 \mid Q_2]$$

$$\equiv [Q_1 \mid U]$$

link $\leftarrow$ $\rightarrow$ unity matrix

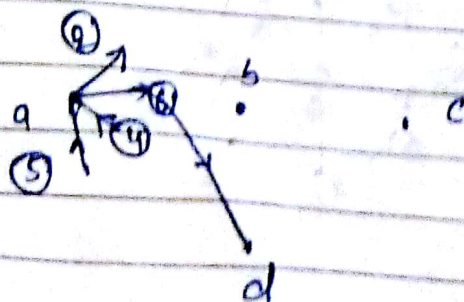
Reduced Incidence matrix of a graph is given by

	1	2	3	4	5	6
a	0	1	0	-1	-1	1
b	0	-1	1	0	1	0
c	1	0	-1	0	0	0

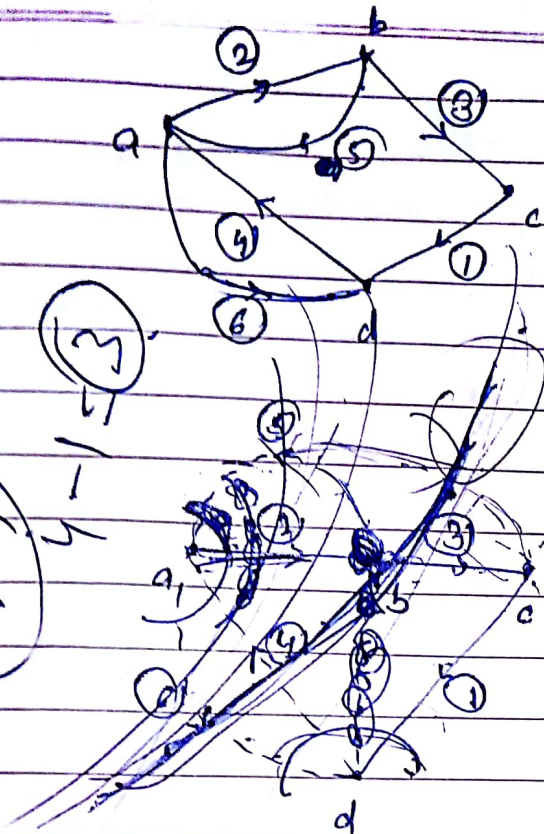
Q. Draw the oriented graph, Select the tree $[1, 3, 1]$ and find f-cut set matrix.

Completing the Incidence matrix

	1	2	3	4	5	6
a	0	1	0	-1	-1	1
b	0	-1	1	0	1	0
c	1	0	-1	0	0	0
d	-1	0	0	1	0	-1



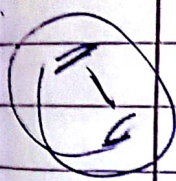
by .
N-1 cut set



$$C_1 = [1, 4, 6]$$

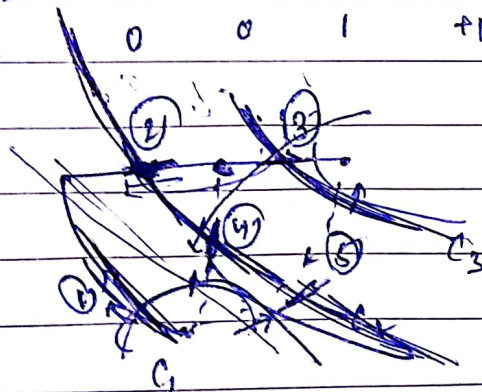
$$C_2 = [2, 4, 5, 6]$$

$$C_3 = [3, 4, 6]$$



	1	2	3	4	5	6
C_1	1	0	0	+1	0	-1
C_2	0	1	0	+1	-1	-1
C_3	0	0	1	+1	0	-1

Q.



$$C_1 = [1, 4, 5]$$

$$C_2 = [2, 4, 5]$$

$$C_3 = [3, 5]$$

	1	2	3	4	5
C_1	1	0	0	+1	-1
C_2	0	1	0	-1	+1
C_3	0	0	1	0	+1